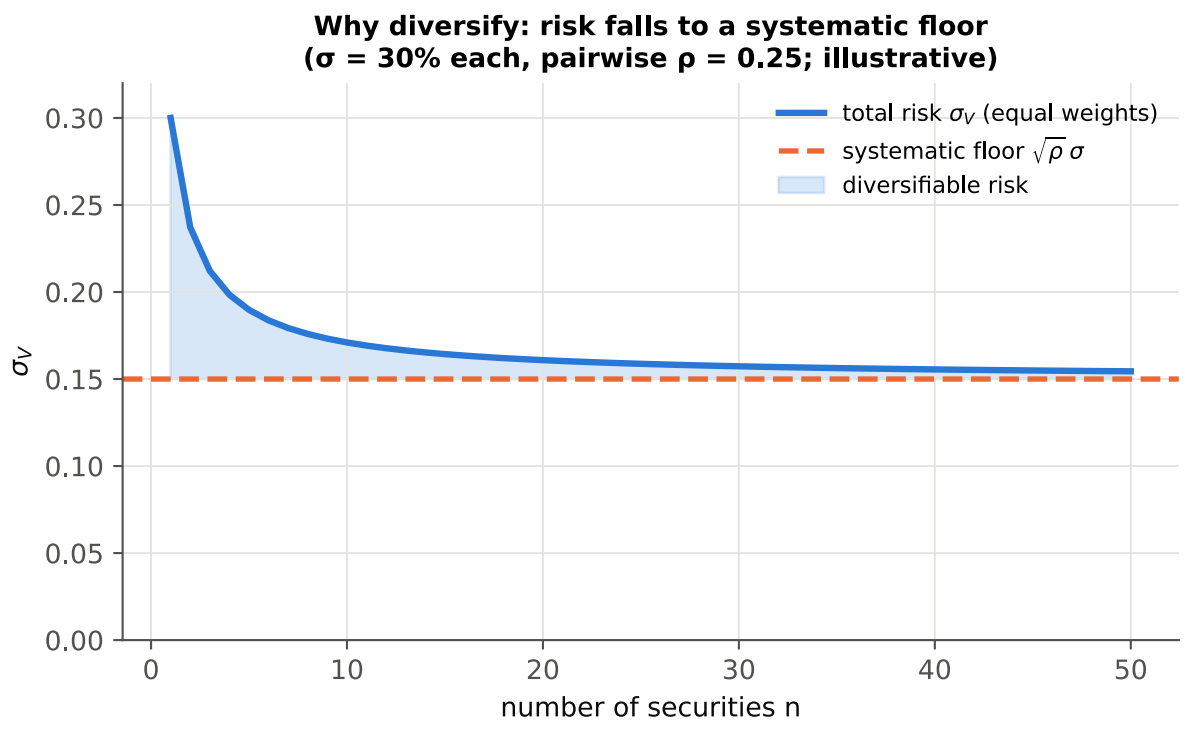
Why diversify: risk falls to a systematic floor
(σ = 30% each, pairwise ρ = 0.25; illustrative)
total risk σ_V (equal weights)
systematic floor $\sqrt{\rho}\,\sigma$
diversifiable risk
σ_V
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0
10
20
30
40
50
number of securities n